$n = 49 \qquad \sigma = 5 \qquad \bar{Y} = 10$

$$CI\left(\bar{Y} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \,,\, \bar{Y} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right)$$

5% confident $\rightarrow z_{0.95/2} = z_{0.475} = 0.06$

$$CI\left(10 - 0.06 \cdot \frac{5}{7} \,,\, 10 + 0.06 \cdot \frac{5}{7}\right)$$

$$CI(10 - 0.04 \,,\, 10 + 0.04)$$

$$CI(9.96 \,,\, 10.04)$$

95% confident $\rightarrow z_{0.05/2} = z_{0.025} = 1.96$

$$CI\left(10 - 1.96 \cdot \frac{5}{7} \,,\, 10 + 1.96 \cdot \frac{5}{7}\right)$$

$$CI(10 - 1.4 \,,\, 10 + 1.4)$$

$$CI(8.6 \,,\, 11.4)$$

99% confident $\rightarrow z_{0.01/2} = z_{0.005} = 2.57$

$$CI\left(10 - 2.57 \cdot \frac{5}{7} \,,\, 10 + 2.57 \cdot \frac{5}{7}\right)$$

$$CI(10 - 1.84 \,,\, 10 + 1.84)$$

$$CI(8.16 \,,\, 11.84)$$